

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	Set B / 2nd year	Date:	15/07/2026

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A programmer wants to understand how an 8085 microprocessor executes different categories of instructions. To examine the execution sequence, write and execute an 8085 program containing data transfer, arithmetic, and branch instructions using an 8085 simulator. Observe the changes in the Program Counter (PC), Stack Pointer (SP), and memory locations during execution. Analyse how different instruction types progress through the fetch, decode, and execute cycle and explain how they influence the execution sequence.
2. A digital electronics engineer is verifying binary patterns used to control hardware devices. To investigate the correctness of the data, use the Windows Calculator (Programmer Mode) to perform bitwise operations (AND, OR, XOR, and NOT) on the given hexadecimal values. Analyse how the hexadecimal and binary representations change after each operation, and determine how bitwise operations are useful in computer systems. Support your analysis with suitable screenshots.
3. A user notices that a computer responds differently when opening a text editor, a media player, and a web browser simultaneously. To examine the execution of these tasks, perform the operations on the computer and observe the interaction between the CPU, memory, and I/O devices. Analyse which functional unit is primarily responsible for processing, storing, and transferring data during each operation, and explain how these components work together to complete the tasks.

Code of Conduct

I U. Vishwa / 192511192 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: U. Vishwa

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

Course Coordinator

Faculty Incharge

Q1. 8085 PROGRAM EXECUTION IN GNUSIM8085

Program: Data transfer, arithmetic and branch instructions

GNU Sim8085

File Edit Assembler Simulator View Help

Editor

01 LXI SP, 3000H
02 MVI A, 05H
03 MVI B, 03H
04 MOV C, A
05 ADD B
06 JNZ NEXT
07 HLT
08
09 NEXT: STA 2500H
10 HLT

Registers

A	05	PC	000A
B	03	SP	3000
C	05	S	0
D	00	Z	0
E	00	AC	0
H	00	P	1
L	00	CY	0

Memory

24F0	00	00	00	00	00	00	00	00
2500	08	00	00	00	00	00	00	00
2510	00	00	00	00	00	00	00	00

Observation:

- After execution, A = 08H
- Result 08H is stored in memory location 2500H
- PC points to the last HLT instruction
- SP remains 3000H

Explanation:

1. MVI A,05H → Load 05H into A
2. MVI B,03H → Load 03H into B
3. MOV C,A → Copy contents of A to C
4. ADD B → Add B to A (05H + 03H = 08H)
5. JNZ NEXT → Since Zero flag = 0, jump to NEXT
6. STA 2500H → Store A (08H) at memory 2500H
7. HLT → Stop execution

Fetch – Decode – Execute Cycle

Cycle	Description	Example (ADD B)
Fetch	CPU fetches the instruction from memory using PC	PC points to ADD B
Decode	Control Unit decodes the instruction	Instruction identified as ADD
Execute	ALU performs addition, result stored in A	A = 08H, Flags updated
Update	PC updated to next instruction	PC = address of next instruction

Observation:

- program starts execution from the first memory location.
- Data transfer instructions load data into registers.
- Arithmetic instruction updates the Accumulator (A)
- Branch instruction changes the program counter (PC)
- Stack pointer (SP) remains unchanged because no stack operation is used.
- Result is stored in memory location 2500H.

Analysis:

- The CPU executes every instruction through Fetch → Decode → Execute cycle.
- Program counter points to the next instruction after execution.
- Registers are updated according to the instruction executed.
- Memory stores the final result permanently

Q2. BITWISE OPERATIONS USING WINDOWS CALCULATOR (PROGRAMMER MODE)

Values taken: A = 5AH B = 3CH

Calculator - [] X

≡ Programmer

1) AND Operation

5A AND 3C = **18**

HEX 18
DEC 24
OCT 30
BIN 0001 1000

QWORD MS

Bit	32	31	0
0000	0000	0000	0000
0000	0000	0001	1000

Calculator - [] X

≡ Programmer

2) OR Operation

5A OR 3C = **7E**

HEX 7E
DEC 126
OCT 176
BIN 0111 1110

QWORD MS

Bit	32	31	0
0000	0000	0000	0000
0000	0000	0111	1110

Calculator - [] X

≡ Programmer

3) XOR Operation

5A XOR 3C = **66**

HEX 66
DEC 102
OCT 146
BIN 0110 0110

QWORD MS

Bit	32	31	0
0000	0000	0000	0000
0000	0010	0110	0110

Calculator - [] X

≡ Programmer

4) NOT Operation (A)

NOT 5A = **A5**

HEX A5
DEC 165
OCT 245
BIN 1010 0101

QWORD MS

Bit	32	31	0
0000	0000	0000	0000
0000	0000	1010	0101

Analysis: Bitwise operations manipulate individual bits of data. These operations are widely used in masking, toggling, data comparison, encryption, and hardware control in computer systems.

Bit wise operations:

Aim:

Perform AND, OR, XOR and NOT operations.

Contents:

AND $\rightarrow$ common bits

OR $\rightarrow$ combined bits

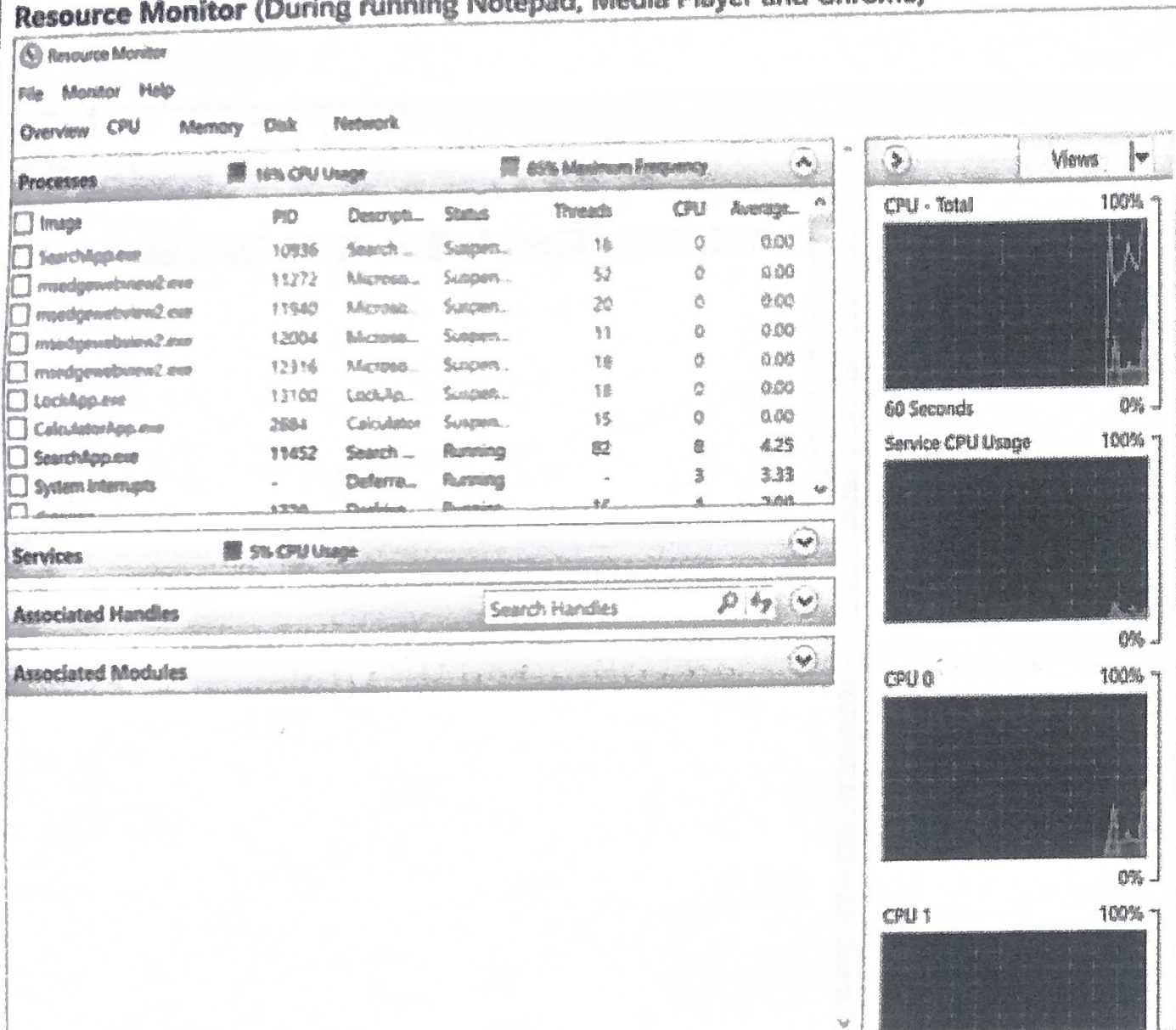
XOR $\rightarrow$ Different bits

NOT $\rightarrow$ Inverts bits

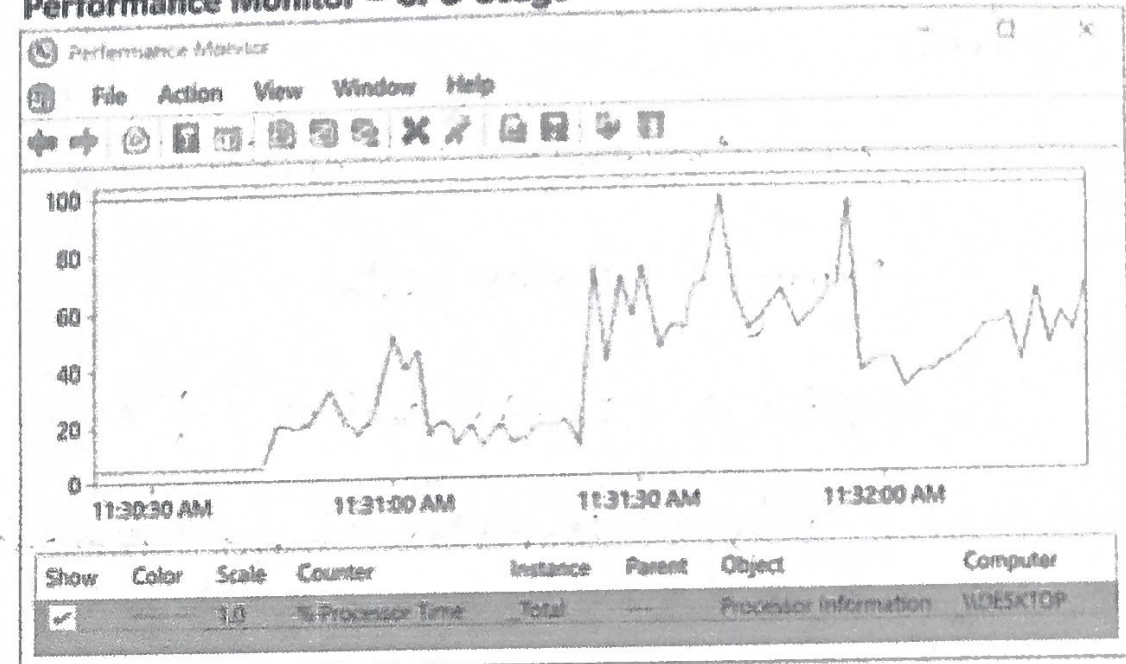
Conclusion: Bit wise operations are used for masking, control, and data processing.

Q3. OBSERVATION USING RESOURCE MONITOR AND PERFORMANCE MONITOR

Resource Monitor (During running Notepad, Media Player and Chrome)



Performance Monitor – CPU Usage



Analysis:

- CPU is the main processing unit that executes instructions of Notepad, Media Player and Chrome.
- Memory stores program code and data temporarily for quick access.
- Disk performs read/write operations to load programs and save files.
- Network handles internet operations for the browser.
- I/O devices (keyboard, mouse, display, speakers) provide input and output to the user.
- All these functional units work together to complete multitasking efficiently.

Aim:

Observe system performance while running applications

Contents:

CPU $\rightarrow$ Processes instructions

Memory $\rightarrow$ stores data.

Disk $\rightarrow$ Reads/writes / files

I/O $\rightarrow$ Handles input/output.

Conclusion:

CPU, memory, Disk, and I/O work together for smooth multitasking.